COURSE-BASED PROJECT DOCUMENTATION | DOC-20

Demo Presentation & Viva Script

Practical Verbal Presentation Script, Live Demo Guide, & Defense Flow

1. 30-Second Elevator Pitch

30-SECOND OPENING PITCH

"Respected examiners, our project is a high-precision Java 21 Syntax Checker. It implements a formal compiler front end combining a GNU Flex lexical DFA, a GNU Bison LALR(1) shift-reduce parser, a native C shared library, and a modern Python Tkinter GUI. It validates modern Java 21 syntax?including record patterns, switch guards, and sealed classes?pinpointing syntax errors to the exact line with a 100% verified test pass rate."

2. 2-Minute Comprehensive Presentation Script

Follow this structured script during your formal project review:

- ? 1. Motivation: Explain that modern Java 21 LTS features (record patterns, switch guards) are rarely covered in academic parsers.
- ? 2. Architecture: Walk through the pipeline from Java Source -> Flex -> Tokens -> Bison LALR(1) -> C DLL -> Python ctypes -> Tkinter GUI.
- ? 3. Grammar Engineering: Highlight how ANTLR4 reference grammar was adapted to LALR(1) left-recursive rules and precedence disambiguation.
- ? 4. Error Precision: Explain how YY_USER_ACTION synchronizes yylloc to capture the exact error line in yyerror().
- ? 5. Results: Announce 59/59 parser tests passed and 14/14 GUI integration tests passed.

3. Step-by-Step Live Demonstration Sequence

Demo Step	Action in GUI / Terminal	Expected Demonstration Outcome
Step 1: Launch GUI	Run: python gui/app.py	Modern dark VS Code theme opens with e...
Step 2: Command Palette	Press Ctrl+Shift+P	Floating overlay opens; type 'Check' a...
Step 3: Valid Java 21	Load SampleValid.java with recor...	Status bar turns green: 'Parsing Succe...
Step 4: Invalid Java	Introduce error: int b = ; on Li...	Status bar turns red: Line 4 highlight...
Step 5: Code Fix	Fix error: int b = 42; and press F5	Error highlight clears instantly; gree...
Step 6: Batch Tests	Run: python tests/run_tests.py	Terminal displays 59/59 PASS (100.0%) ...

4. Concluding Script

CLOSING STATEMENT

"To conclude, our system proves that formal compiler theory can be cleanly translated into a high-performance, modern developer tool. All source code, grammar rules, native libraries, and tests are complete and verified. Thank you, and I am happy to answer any questions."